GIT & GITHUB

Version Control · Collaboration · Deployment

A complete technical guide to modern version control

v2.43+ | GitHub Desktop | Static Deployment

What is Git?

The world's most popular version control system

70%+

developers use Git

Created by	Linus Torvalds in 2005
Maintained by	Junio Hamano
Type	Distributed Version Control System
Host	GitHub (owned by Microsoft since 2018)

Key Git Concepts

Repository

Folder where Git tracks your project and history

Branch

Work on different versions or features simultaneously

Clone

Make a copy of a remote repo on your computer

Merge

Combine changes from different branches

Stage

Tell Git which changes you want to save next

Pull

Get the latest changes from a remote repository

Commit

Save a permanent snapshot of staged changes

Push

Send your local changes to a remote repository

Working with Git

1

Initialize

Run git init to make a folder a Repository — Git creates a hidden .git folder

2

Modify

Create, edit or delete files — Git marks them as modified

3

Stage

git add <file> — select the changes you want in your next snapshot

4

Commit

git commit — Git stores a permanent snapshot with a message

5

History

git log — view full history of every commit and revert anytime

💡 Most Git actions happen locally. Only Push and Pull interact with remote servers like GitHub, GitLab or Bitbucket.

Why Git?



Global Collaboration

Developers work together from anywhere in the world



Full History

See every change ever made — who, when and why



Time Travel

Revert to any previous commit in seconds



Efficient Storage

Tracks changes, not full file copies each commit

Git vs GitHub

Git

- Command-line tool
- Runs on your computer
- Tracks changes locally
- Free & open source
- No internet needed

Est. 2005 · Linus Torvalds

GitHub

- Cloud hosting platform
- Runs in the browser
- Hosts remote repos
- Owned by Microsoft
- Push/Pull over internet

Est. 2008 · Microsoft (since 2018)

What is GitHub?

The world's largest host of source code — built on Git

100M+

developers worldwide

What it is	A cloud platform that hosts Git repositories online
Relationship	GitHub makes tools that USE Git — they are not the same thing
Owner	Acquired by Microsoft in 2018
Founded	2008 — started as a startup, now the industry standard



Repositories



Pull Requests



Issues



GitHub Actions



GitHub Pages

What is Git Bash?

A terminal for Windows that lets you run Git commands using Unix-style shell



Windows Only

Installed automatically with Git for Windows from git-scm.com



Unix Commands

Supports bash commands like ls, pwd, cd, mkdir, rm alongside git



Full Git Support

Run any git command — init, clone, commit, push, pull and more



Start Menu

Find Git Bash in Start → type "Git Bash" → open the app

Installing Git

Download free from git-scm.com — includes Git Bash on Windows

Windows

1. Download installer from git-scm.com
2. Run .exe and click "Next"
3. Git + Git Bash installed

macOS

1. brew install git (Homebrew)
2. Or download the .dmg file
3. Drag Git to Applications

Linux (Ubuntu)

1. sudo apt-get install git
2. Enter password when prompted
3. Ready to use immediately

bash · verify installation

```
$ git --version
git version 2.43.0.windows.1

$ # You are ready to go!
```

GitHub Desktop

A GUI client for Git that simplifies version control

Visual Interface

See changes, diffs and history without memorizing commands

Find Commands

Access less-common Git features easily from the GUI

Best Practices

Encourages clean commit history and well-structured messages

GitHub Integration

Authenticate and manage PRs without leaving the app

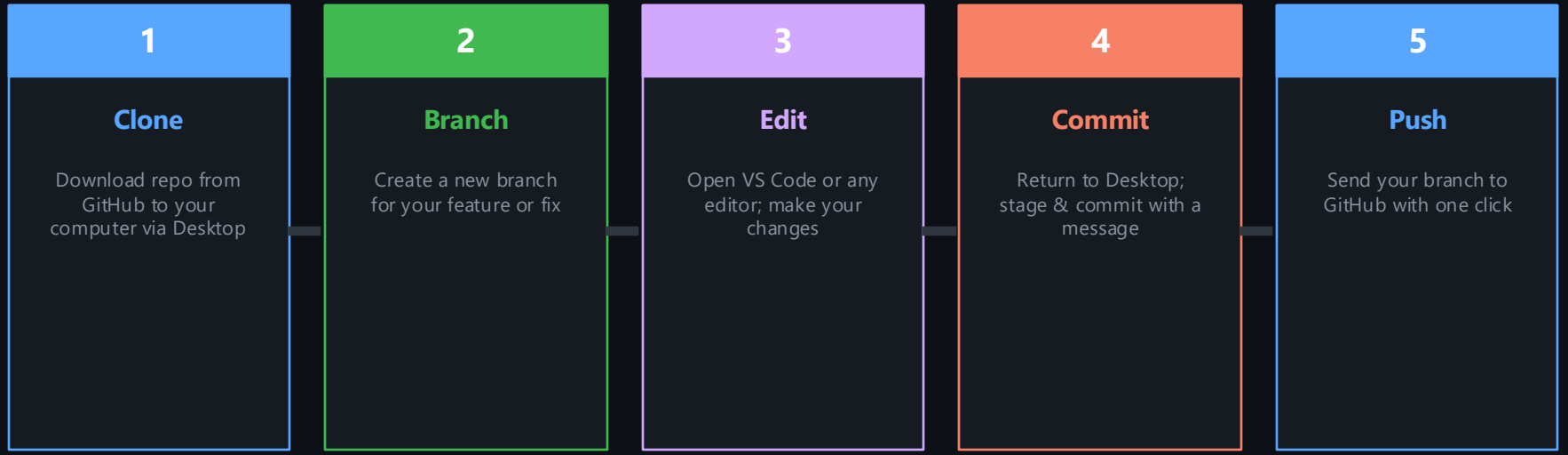
Branch Easily

Create, switch and merge branches with a single click

Co-author Commits

Credit collaborators directly from the commit dialog

Typical GitHub Desktop Workflow



💡 GitHub Desktop works alongside any editor (VS Code, Sublime, etc.).
You can authenticate to GitHub without a separate credential manager.
Check out pull requests and run CI checks directly from the Desktop app.

Essential Git Commands

`git init`	Initialize a new local repository
`git clone <url>`	Clone a remote repository
`git add <file>`	Stage a specific file for commit
`git add .`	Stage all modified files
`git commit -m "msg"`	Commit staged changes with message
`git status`	Show working tree status
`git log`	Show commit history
`git branch <name>`	Create a new branch
`git checkout <name>`	Switch to a branch
`git merge <name>`	Merge a branch into current
`git pull`	Fetch and merge from remote
`git push`	Push local commits to remote

Deploy Static HTML with GitHub Pages

Host your website for FREE directly from a GitHub repository

01

Create Repo

Create a new public GitHub repository

02

Add Files

Add your index.html, CSS & JS files and push

03

Settings

Go to Settings → Pages in your repository

04

Enable Pages

Set Source branch (main) and folder (/root)

05

Live URL

Your site is live at username.github.io/repo

GitHub Pages URL format:

`https://username.github.io/repository-name`

Supports: HTML · CSS · JavaScript · React (with build) · Vue (with build) · Jekyll (natively)

Deploying — The Commands

bash · full deploy workflow

```
# 1. Initialize & connect to GitHub
git init
git remote add origin https://github.com/username/my-site.git

# 2. Add your static files and commit
git add .
git commit -m "Initial site commit"

# 3. Push to GitHub
git push -u origin main

# 4. Enable GitHub Pages in repo Settings → Pages
#    Set branch: main  |  folder: / (root)

# 5. Your site is LIVE at:
#    https://username.github.io/my-site
```

Summary

Git	A free, open-source distributed version control system
GitHub	The world's largest host of Git repositories (Microsoft)
Key Actions	init, clone, add, commit, push, pull, merge, branch
GitHub Desktop	GUI client — visual, beginner-friendly, full-featured
GitHub Pages	Free static site hosting from any public repository
Deployment	Push HTML/CSS/JS → enable Pages → instant live URL